

Ahmadreza Farahani | ML/AI Software Engineer

📧 amadrezafrh.github.io | ✉ amadreza.farahani@outlook.com | 🏠 Turin, Italy (open to relocation)
🌐 linkedin.com/in/amadrezafrh | ☎ +39 351 641 1678 | 📅 4+ Years of Experience

Education

M.Sc. Computer Engineering - Artificial Intelligence & Data Analytics

Politecnico di Torino

Turin, Italy

2022 - 2025

- **Courses:** Advanced Machine Learning, System and Device Programming, Software Engineering, Web Applications
- **Thesis:** RobVC: A Robust zero-shot End-to-End Self Supervised Voice Conversion (Funded by CROSSING Co. in Darmstadt) | [Blog](#)
- **GPA:** 3.42/4.00

B.Sc. Electrical and Computer Engineering - Telecommunications

Iran University of Science and Technology

Tehran, Iran

2016 - 2021

- **Courses:** Programming, Digital Signal Processing, Linear Algebra, Linear Control Systems, Communications
- **Thesis:** Deep Learning-Based Vehicle Detection and Traffic Flow Analysis (Funded by Megamouj Co.) | [Blog](#)
- **GPA:** 2.96/4.00

Professional Experience

AI/ML Software Engineer

Zirak

Turin, Italy

11/2024 - Current

- Led development of a real-time meeting assistant agent with a **LangGraph** human-in-the-loop moderator and episodic memory (**RAG**), hitting sub-2s latency and an 83% cut in repeated output verified via **LangSmith** evals (golden eval set).
- Automated 81% of inbound emails fully on-premise by developing an email observer agent for Traco using fine-tuned Llama 3.2 3B (**QLoRA**) and **LangChain** that achieved a 94% F1 on structured extraction and routed low-confidence cases to human review.
- Owned and delivered a multi-hop QA agent on **AWS Bedrock**, choosing **GraphRAG** over flat RAG to query over 1,000 linked documents that achieved 93% tool-call accuracy.
- Drove the architecture of an ISO document change-analysis product detecting 96% of revision changes, with a hybrid **BM25** and a dense retrieval pipeline into **pgvector** alongside model routing that cut LLM costs by 65% without losing recall.
- Reduced query SQL authoring time by 70% for non-technical analysts at Poste Italiane's logistics division with a production NL-to-SQL agent: **LangGraph** routing **MCP** tools across PostgreSQL and MySQL, served on-prem via FastAPI and **Ollama**.
- Spearheaded a document-review agent (HuggingFace models served on **vLLM** with **Guardrails** and local **ChromaDB** vector storage), saving 50 hours/month internally and later licensed as SaaS to external clients like Techyon.
- Designed a rail-safety detection and segmentation pipeline for Blue Engineering selected YOLOv8 over the existing YOLOv5x system, that improved mAP by 3.3% , deployed on **AWS ECS** with Prometheus and Grafana for real-time monitoring.

Visiting Researcher - Master Thesis

Technische Universität Darmstadt | CROSSING

Darmstadt, Germany

09/2023 - 11/2024

- Architected a Diffusion **Transformer** for voice conversion (Python, PyTorch) on HPC GPUs, improving speaker similarity by 9% over open-source baselines with fewer sampling steps, enabling real-time conversion | [Blog](#)
- Ran a prototype low-latency inference on AWS ECS, tracking experiments in **MLflow** and monitoring latency with CloudWatch.

ML Software Engineer

PartAI Research Center

Tehran, Iran

02/2020 - 03/2022

- Optimized an on-device deep neural network keyword-spotting, with hard negative samples to reduce false positives by 30%.
- Fine-tuned internal speaker verification system on new domains in **TensorFlow** that reduced the Equal Error Rate (EER) by 2%.
- Established Test-Driven Development (TDD) for Persian ASR and **NLP** services, wrote unit/integration tests reaching 100% coverage and set up **Jenkins CI/CD** pipelines.

Internship Experience

Associate Generative AI Software Engineer

AROL S.p.A

Canelli, Italy

09/2022 - 04/2023

- Developed a synthetic machinery data generator (**GAN**, TensorFlow, trained on **AWS SageMaker**) packaged with MongoDB and **Docker** Compose plus REST API docs, so customers run it locally without exposing proprietary data.
- Defined KPIs and **Python** tests validating synthetic vs. real machinery signals and API responses, wired into the team's CI pipeline to replace manual QA before customer demos.

Computer Vision Research Intern - Bachelor Thesis

Megamouj Co. | Tehran University of Science and Technology

Tehran, Iran

09/2019 - 02/2020

- Automated a traffic-flow monitoring system (C++, Python, **OpenCV**, YOLOv3) for vehicle detection, tracking, and flow estimation, reducing manual video review in a national-scale project | [Blog](#)
- Engineered a traffic-flow dataset (**NumPy**, Pandas, **scikit-learn**) to fine-tune the existing object detection model, changing the precision from 45.2% to 72%.

Skills

Agentic AI	LangGraph, LangChain, LangSmith, MCP, agentic RAG (hybrid retrieval, reranking), guardrails, Pydantic
LLM Systems	Transformers, HuggingFace, Ollama, vLLM, TorchServe, Milvus, ChromaDB, pgvector, LoRA, QLoRA
Machine Learning	NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, XGBoost, MLflow, GANs, U-Net, YOLO, ASR
Cloud Technologies	AWS (Bedrock, ECS, EKS, EC2, S3, Lambda, SageMaker, CloudWatch), Docker
Software Development	Python, C/C++, SQL, Bash, Linux, Git, CI/CD, FastAPI, Prometheus, Grafana,